Example: Bayes filter
Reference: Thrun, Burgard, Fox, "Probabilistic Robotics," MIT Press 2006

1: **Algorithm Bayes_filter**($bel(x_{t-1}), u_t, z_t$)**:**
2: for all x_t do
3: $\overline{bel}(x_t) = \int p(x_t \mid u_t, x_{t-1})\ bel(x_{t-1})\ dx$
4: $bel(x_t) = \eta\ p(z_t \mid x_t)\ \overline{bel}(x_t)$
5: endfor
6: return $bel(x_t)$

Table 2.1 The general algorithm for Bayes filtering.

expressed by non-uniform distributions; however, the two cases of full knowledge and full ignorance are the most common ones in practice.

The algorithm Bayes filter can only be implemented in the form stated here for very simple estimation problems. In particular, we either need to be able to carry out the integration in Line 3 and the multiplication in Line 4 in closed form, or we need to restrict ourselves to finite state spaces, so that the integral in Line 3 becomes a (finite) sum.

2.4.2 Example

Our illustration of the Bayes filter algorithm is based on the scenario in Figure 2.2, which shows a robot estimating the state of a door using its camera. To make this problem simple, let us assume that the door can be in one of two possible states, open or closed, and that only the robot can change the state of the door. Let us furthermore assume that the robot does not know the state of the door initially. Instead, it assigns equal prior probability to the two possible door states:

$$bel(X_0 = \mathbf{open}) = 0.5 \tag{2.37}$$
$$bel(X_0 = \mathbf{closed}) = 0.5 \tag{2.38}$$

Let us furthermore assume the robot's sensors are noisy. The noise is characterized by the following conditional probabilities:

$$\begin{aligned} p(Z_t = \mathbf{sense_open} \mid X_t = \mathbf{is_open}) &= 0.6 \\ p(Z_t = \mathbf{sense_closed} \mid X_t = \mathbf{is_open}) &= 0.4 \end{aligned} \tag{2.39}$$

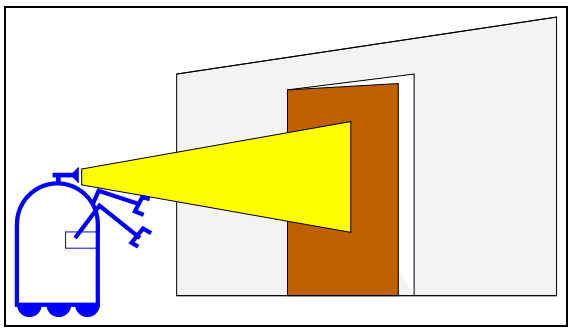


Figure 2.2 A mobile robot estimating the state of a door.

and

$$\begin{aligned} p(Z_t = \textbf{sense_open} \mid X_t = \textbf{is_closed}) &= 0.2 \\ p(Z_t = \textbf{sense_closed} \mid X_t = \textbf{is_closed}) &= 0.8 \end{aligned} \tag{2.40}$$

These probabilities suggest that the robot's sensors are relatively reliable in detecting a *closed* door, in that the error probability is 0.2. However, when the door is open, it has a 0.4 probability of a false measurement.

Finally, let us assume the robot uses its manipulator to push the door open. If the door is already open, it will remain open. It it is closed, the robot has a 0.8 chance that it will be open afterwards:

$$p(X_t = \textbf{is_open} \mid U_t = \textbf{push}, X_{t_1} = \textbf{is_open}) = 1$$

$$p(X_t = \textbf{is_closed} \mid U_t = \textbf{push}, X_{t_1} = \textbf{is_open}) = 0 \tag{2.41}$$

$$p(X_t = \textbf{is_open} \mid U_t = \textbf{push}, X_{t_1} = \textbf{is_closed}) = 0.8$$

$$p(X_t = \textbf{is_closed} \mid U_t = \textbf{push}, X_{t_1} = \textbf{is_closed}) = 0.2 \tag{2.42}$$

It can also choose not to use its manipulator, in which case the state of the world does not change. This is stated by the following conditional probabilities:

$$p(X_t = \textbf{is_open} \mid U_t = \textbf{do_nothing}, X_{t_1} = \textbf{is_open}) = 1$$

$$p(X_t = \textbf{is_closed} \mid U_t = \textbf{do_nothing}, X_{t_1} = \textbf{is_open}) = 0 \tag{2.43}$$

$$p(X_t = \textbf{is_open} \mid U_t = \textbf{do_nothing}, X_{t_1} = \textbf{is_closed}) = 0$$

$$p(X_t = \textbf{is_closed} \mid U_t = \textbf{do_nothing}, X_{t_1} = \textbf{is_closed}) = 1 \tag{2.44}$$

Suppose at time t, the robot takes no control action but it senses an open door. The resulting posterior belief is calculated by the Bayes filter using the prior belief $bel(X_0)$, the control $u_1 = \textbf{do_nothing}$, and the measurement $\textbf{sense_open}$ as input. Since the state space is finite, the integral in Line 3 turns into a finite sum:

$$\begin{aligned}
\overline{bel}(x_1) &= \int p(x_1 \mid u_1, x_0)\, bel(x_0)\, dx_0 \\
&= \sum_{x_0} p(x_1 \mid u_1, x_0)\, bel(x_0) \\
&= p(x_1 \mid U_1 = \textbf{do_nothing}, X_0 = \textbf{is_open})\, bel(X_0 = \textbf{is_open}) \\
&\quad + p(x_1 \mid U_1 = \textbf{do_nothing}, X_0 = \textbf{is_closed})\, bel(X_0 = \textbf{is_closed})
\end{aligned} \tag{2.45}$$

We can now substitute the two possible values for the state variable X_1. For the hypothesis $X_1 = \textbf{is_open}$, we obtain

$$\begin{aligned}
&\overline{bel}(X_1 = \textbf{is_open}) \\
&= p(X_1 = \textbf{is_open} \mid U_1 = \textbf{do_nothing}, X_0 = \textbf{is_open})\, bel(X_0 = \textbf{is_open}) \\
&\quad + p(X_1 = \textbf{is_open} \mid U_1 = \textbf{do_nothing}, X_0 = \textbf{is_closed})\, bel(X_0 = \textbf{is_closed}) \\
&= 1 \cdot 0.5 + 0 \cdot 0.5 \;=\; 0.5
\end{aligned} \tag{2.46}$$

Likewise, for $X_1 = \textbf{is_closed}$ we get

$$\begin{aligned}
&\overline{bel}(X_1 = \textbf{is_closed}) \\
&= p(X_1 = \textbf{is_closed} \mid U_1 = \textbf{do_nothing}, X_0 = \textbf{is_open})\, bel(X_0 = \textbf{is_open}) \\
&\quad + p(X_1 = \textbf{is_closed} \mid U_1 = \textbf{do_nothing}, X_0 = \textbf{is_closed})\, bel(X_0 = \textbf{is_closed}) \\
&= 0 \cdot 0.5 + 1 \cdot 0.5 \;=\; 0.5
\end{aligned} \tag{2.47}$$

The fact that the belief $\overline{bel}(x_1)$ equals our prior belief $bel(x_0)$ should not surprise, as the action $\textbf{do_nothing}$ does not affect the state of the world; neither does the world change over time by itself in our example.

Incorporating the measurement, however, changes the belief. Line 4 of the Bayes filter algorithm implies

$$bel(x_1) = \eta\, p(Z_1 = \textbf{sense_open} \mid x_1)\, \overline{bel}(x_1)\,. \tag{2.48}$$

For the two possible cases, $X_1 = \textbf{is_open}$ and $X_1 = \textbf{is_closed}$, we get

$$\begin{aligned} & bel(X_1 = \textbf{is_open}) \\ &= \eta\, p(Z_1 = \textbf{sense_open} \mid X_1 = \textbf{is_open})\, \overline{bel}(X_1 = \textbf{is_open}) \\ &= \eta\, 0.6 \cdot 0.5 \;=\; \eta\, 0.3 \end{aligned} \tag{2.49}$$

and

$$\begin{aligned} & bel(X_1 = \textbf{is_closed}) \\ &= \eta\, p(Z_1 = \textbf{sense_open} \mid X_1 = \textbf{is_closed})\, \overline{bel}(X_1 = \textbf{is_closed}) \\ &= \eta\, 0.2 \cdot 0.5 \;=\; \eta\, 0.1 \end{aligned} \tag{2.50}$$

The normalizer η is now easily calculated:

$$\eta \;=\; (0.3 + 0.1)^{-1} \;=\; 2.5 \tag{2.51}$$

Hence, we have

$$\begin{aligned} bel(X_1 = \textbf{is_open}) &= 0.75 \\ bel(X_1 = \textbf{is_closed}) &= 0.25 \end{aligned} \tag{2.52}$$

This calculation is now easily iterated for the next time step. As the reader easily verifies, for $u_2 = \textbf{push}$ and $z_2 = \textbf{sense_open}$ we get

$$\begin{aligned} \overline{bel}(X_2 = \textbf{is_open}) &= 1 \cdot 0.75 + 0.8 \cdot 0.25 \;=\; 0.95 \\ \overline{bel}(X_2 = \textbf{is_closed}) &= 0 \cdot 0.75 + 0.2 \cdot 0.25 \;=\; 0.05\,, \end{aligned} \tag{2.53}$$

and

$$\begin{aligned} bel(X_2 = \textbf{is_open}) &= \eta\, 0.6 \cdot 0.95 \;\approx\; 0.983 \\ bel(X_2 = \textbf{is_closed}) &= \eta\, 0.2 \cdot 0.05 \;\approx\; 0.017\,. \end{aligned} \tag{2.54}$$

At this point, the robot believes that with 0.983 probability the door is open, hence both its measurements were correct. At first glance, this probability may appear to be

sufficiently high to simply accept this hypothesis as the world state and act accordingly. However, such an approach may result in unnecessarily high costs. If mistaking a closed door for an open one incurs costs (e.g., the robot crashes into a door), considering both hypotheses in the decision making process will be essential, as unlikely as one of them may be. Just imagine flying an aircraft on auto pilot with a perceived chance of 0.983 for not crashing!

2.4.3 Mathematical Derivation of the Bayes Filter

The correctness of the Bayes filter algorithm is shown by induction. To do so, we need to show that it correctly calculates the posterior distribution $p(x_t \mid z_{1:t}, u_{1:t})$ from the corresponding posterior one time step earlier, $p(x_{t-1} \mid z_{1:t-1}, u_{1:t-1})$. The correctness follows then by induction under the assumption that we correctly initialized the prior belief $bel(x_0)$ at time $t = 0$.

Our derivation requires that the state x_t is complete, as defined in Section 2.3.1, and it requires that controls are chosen at random. The first step of our derivation involves the application of Bayes rule (2.23) to the target posterior:

$$\begin{aligned} p(x_t \mid z_{1:t}, u_{1:t}) &= \frac{p(z_t \mid x_t, z_{1:t-1}, u_{1:t})\, p(x_t \mid z_{1:t-1}, u_{1:t})}{p(z_t \mid z_{1:t-1}, u_{1:t})} \\ &= \eta\, p(z_t \mid x_t, z_{1:t-1}, u_{1:t})\, p(x_t \mid z_{1:t-1}, u_{1:t}) \end{aligned} \tag{2.55}$$

We now exploit the assumption that our state is complete. In Section 2.3.1, we defined a state x_t to be complete if no variables prior to x_t may influence the stochastic evolution of future states. In particular, if we (hypothetically) knew the state x_t and were interested in predicting the measurement z_t, no past measurement or control would provide us additional information. In mathematical terms, this is expressed by the following conditional independence:

$$p(z_t \mid x_t, z_{1:t-1}, u_{1:t}) = p(z_t \mid x_t)\,. \tag{2.56}$$

Such a statement is another example of *conditional independence*. It allows us to simplify (2.55) as follows:

$$p(x_t \mid z_{1:t}, u_{1:t}) = \eta\, p(z_t \mid x_t)\, p(x_t \mid z_{1:t-1}, u_{1:t}) \tag{2.57}$$